



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

Subject : SECP3213 BUSINESS INTELLIGENCE

Semester : 2/2526

Section : 02

Task : Business Intelligence Group Project

Phase : Implementation and Data Preparation (Stage 2)

Dataset Link : <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce/data>

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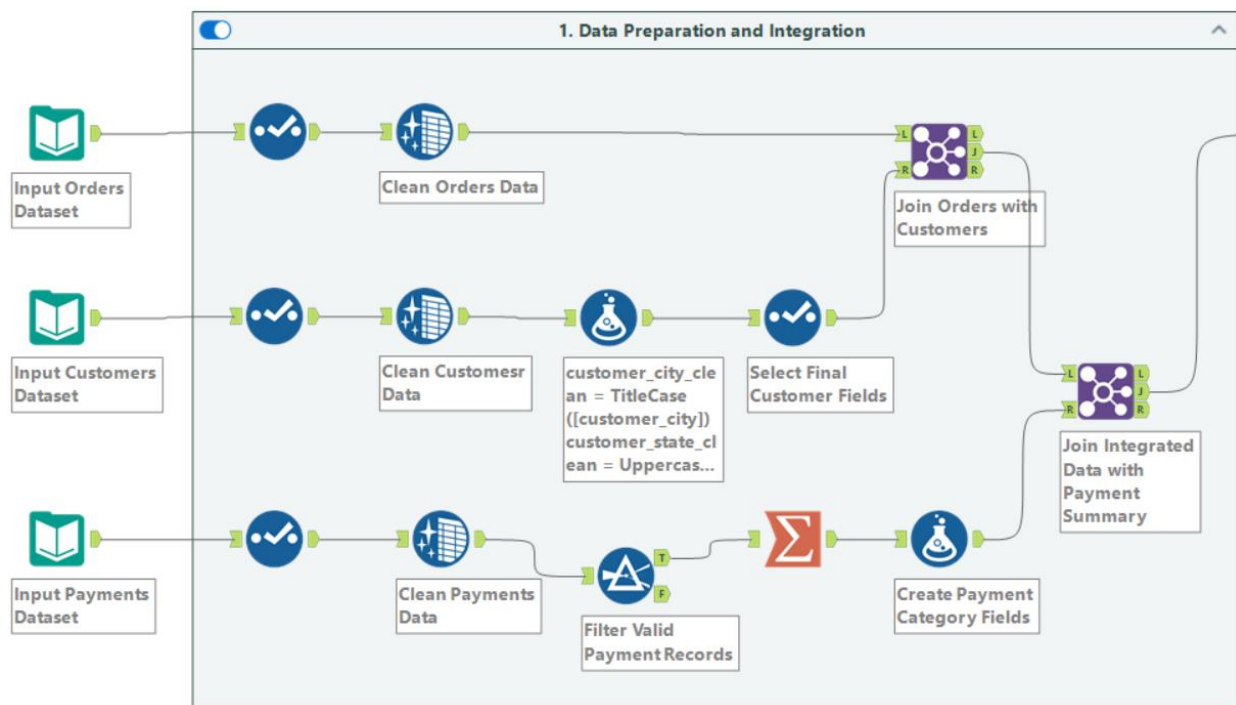
1.0 OVERVIEW

This phase executes the Extract, Transform, Load (ETL) pipeline proposed in Phase 1. Using Alteryx Designer, we process the raw Olist datasets into a single, clean CSV ready for Power BI dashboards.

2.0 Alteryx ETL Workflow

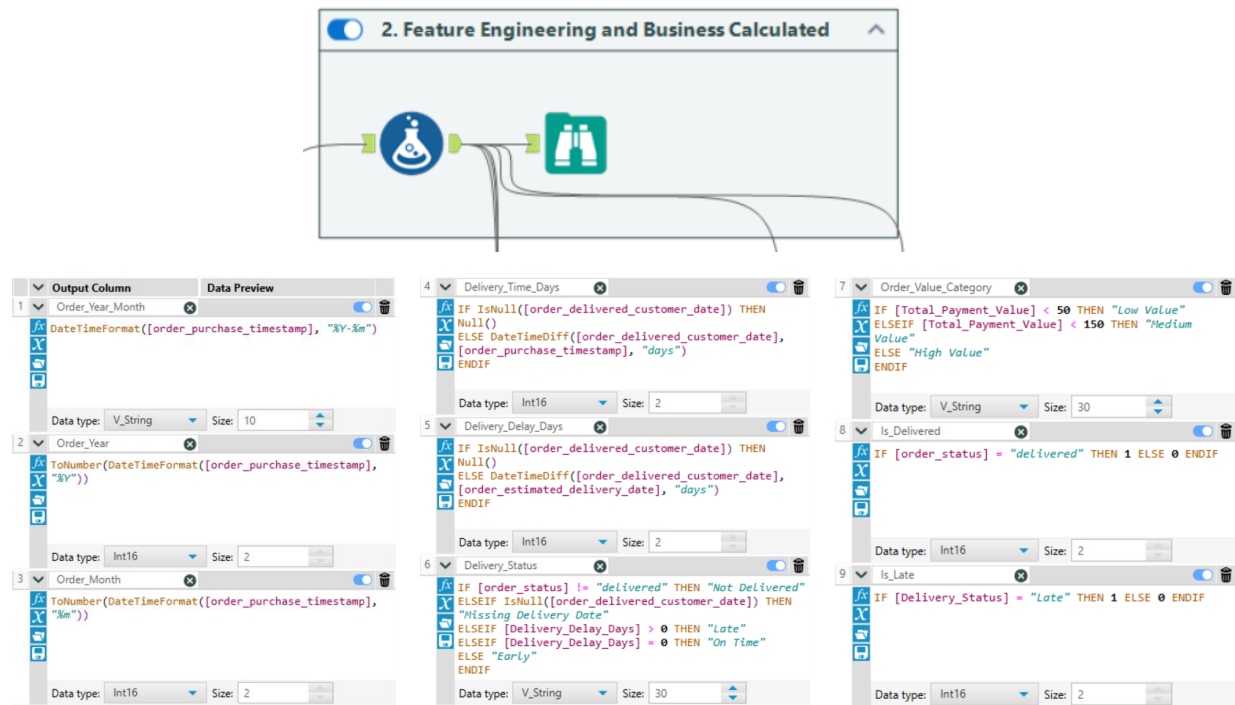
The workflow is divided into four main containers to ensure structured data processing.

2.1 Data Preparation and Integration



Data Extraction	Import the raw CSVs which are Orders, Customers, and Payments datasets
Data Cleansing	Remove nulls and whitespace
▪ Orders	order_id, customer_id, order_status
▪ Customers	customer_id, customer_unique_id, customer_zip_code_prefix, customer_city, customer_state
▪ Payments	order_id, payment_type
Formatting	Standardises customer_city to TitleCase and customer_state to Uppercase
Payment Summarisation	Filter isolates valid payments. Summarize tool then aggregates payment values by order_id to prevent duplicated revenue totals
Data Integration	Join tools consolidate the datasets. Orders + Customers via customer_id, then + Payments via order_id

2.2 Feature Engineering



Part 1 Date and Time Extraction

Order_Year_Month	Extracts year and month for trend analysis
Order_Year	Isolates the year as a number
Order_Month	Isolates the month as a number

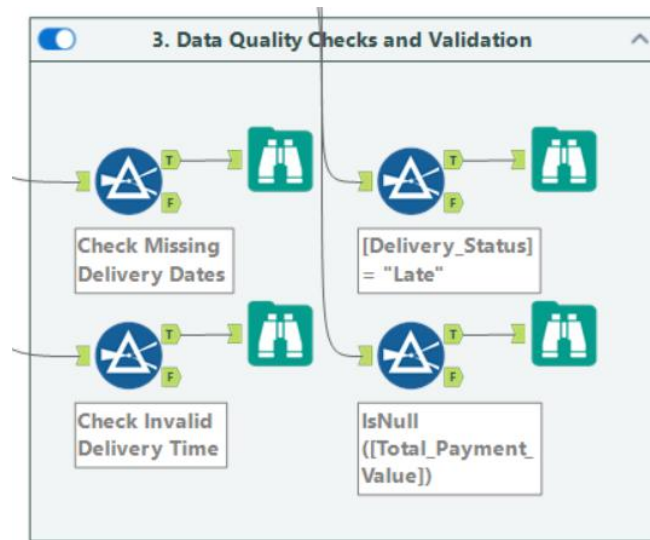
Part 2 Delivery Performance Metrics

Delivery_Time_Days	Calculates the actual days taken to deliver the order
Delivery_Delay_Days	Calculates if an order was delayed past the estimated date
Delivery_Status	Categorises delivery performance (Early, On Time, Late, Missing, or Not Delivered) based on the delay calculation and order status

Part 3 Business Categorisation

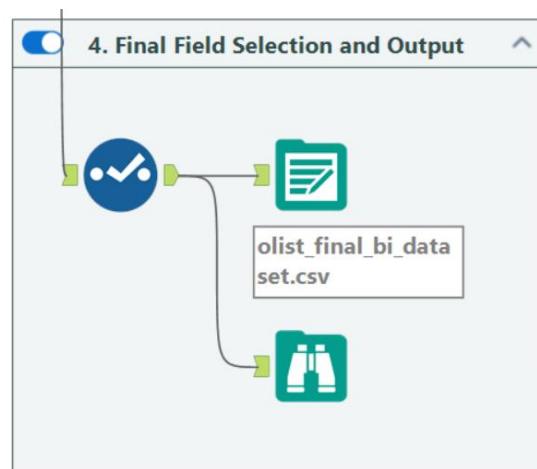
Order_Value_Category	Segments orders into "Low Value" (< 50), "Medium Value" (< 150), and "High Value" (≥ 150) based on Total_Payment_Value
Is_Delivered	Creates a binary flag (1 = delivered, 0 = other) to easily calculate delivery rates
Is_Late	Creates a binary flag (1 = late, 0 = other) to easily calculate late delivery percentages

2.3 Data Quality Checks



Validations	Multiple Browse tools are connected to the workflow after feature engineering.
Purpose	Allows the team to inspect the data, verify that joins worked correctly without data loss, and ensure all mathematical formulas return accurate results before generating the final file.

2.4 Final Output



Trimming	Select tool removes redundant foreign keys, the duplicate order_id columns, and verifies optimal data types
Export	The Output Data tool generates olist_final_bi_dataset.csv as the centralised source for Power BI

3.0 Overall Workflow

